

SHARIF UNIVERSITY OF TECHNOLOGY

Department of Electrical Engineering

PROJECT ASSIGNMENT

MIDI Digital Synthesizer and Audio Effects Prototype

INSTRUCTOR / INSTRUCTORS

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ACADEMIC CONTEXT

Signals and Systems

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1 Deliverables and assessment

Build a MIDI-to-audio synthesizer followed by a configurable series of audio effects, using Python or MATLAB. Package source, report, figures, and audio outputs together. The original naming convention is Project-StudentNo1_StudentNo2_StudentNo3.zip. Supply the report PDF and its LaTeX source when using LaTeX. The course submission requires a GitHub repository. The grading table assigns synthesis 0.45 plus 0.1 bonus, GitHub 0.05, LaTeX/report structure 0.2, and effects 1.2. The stated compulsory/bonus cap applies to the course score. It is not necessary to implement every listed effect to reach the maximum. Keep implemented and proposed features distinguishable.

Table 1: Deliverables and assessment - numerical data and parameters.

Item	**Total Points**
Synthesis	0.45 + 0.1
GitHub	0.05
LATEX+ Report Structure	0.2
Effects	1.2

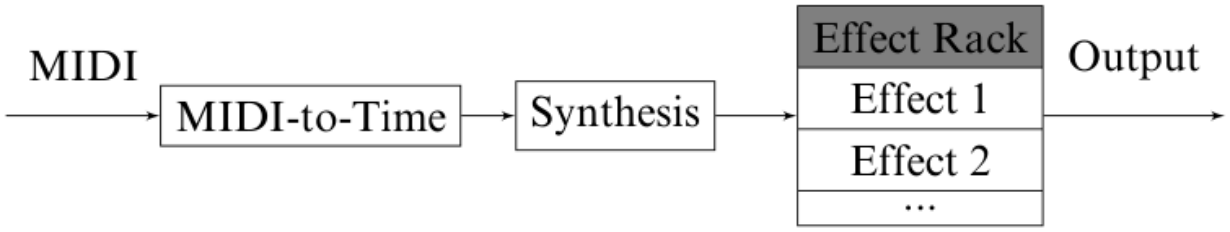


Figure 1: Diagram or recorded result: Deliverables and assessment.

2 Notes, rhythm, and MIDI timing

MIDI describes musical events rather than recorded waveform samples. Equal temperament uses a frequency ratio of 2^(1/12) per semitone, with A4 = 440 Hz. For MIDI note n, f(n) = 440*2^((n-69)/12) Hz. Thus A3 = 220 Hz and A5 = 880 Hz. A periodic tone can contain harmonics at integer multiples of its fundamental; the fundamental need not be the largest spectral coefficient. The tables' MIDI CC heading refers here to note numbers, not control-change identifiers. Read note_on and note_off messages, channel, note number, velocity, and delta time. Channels share a common timeline and must be mixed rather than concatenated. A note_on with zero velocity is ordinarily a note-off event. Pair events by channel and note and define a policy for repeated overlapping notes. Tempo is measured in microseconds per quarter note, and the file specifies ticks per quarter note. Convert delta ticks using the tempo active during that interval. The example tempo 576923, resolution 192 ticks/quarter, and duration 96 ticks give 288461.5 microseconds, approximately 0.288 seconds. A tempo change requires segment-wise conversion.

Table 2: Notes, rhythm, and MIDI timing - numerical data and parameters. (panel 1)

Note	Octave -1	Octave 0	Octave 1	Octave 2	Octave 3	Octave 4
C	8.18	16.35	32.70	65.41	130.81	261.63
C#/Db	8.66	17.32	34.65	69.30	138.59	277.18
D	9.18	18.35	36.71	73.42	146.83	293.66
D#/Eb	9.72	19.45	38.89	77.78	155.56	311.13
E	10.30	20.60	41.20	82.41	164.81	329.63
F	10.91	21.83	43.65	87.31	174.61	349.23
F#/Gb	11.56	23.12	46.25	92.50	185.00	369.99
G	12.25	24.50	49.00	98.00	196.00	392.00
G#/Ab	12.98	25.96	51.91	103.83	207.65	415.30
A	13.75	27.50	55.00	110.00	220.00	440.00
A#/Bb	14.57	29.14	58.27	116.54	233.08	466.16
B	15.43	30.87	61.74	123.47	246.94	493.88

Table 3: Notes, rhythm, and MIDI timing - numerical data and parameters. (panel 2)

Note	Octave 5	Octave 6	Octave 7	Octave 8	Octave 9	Octave 10
C	523.25	1046.50	2093.00	4186.01	8372.02	16744.04
C#/Db	554.37	1108.73	2217.46	4434.92	8869.84	17739.69
D	587.33	1174.66	2349.32	4698.64	9397.27	18794.55
D#/Eb	622.25	1244.51	2489.02	4978.03	9956.06	19912.13
E	659.26	1318.51	2637.02	5274.04	10548.08	21096.16
F	698.46	1396.91	2793.83	5587.65	11175.30	22350.61
F#/Gb	739.99	1479.98	2959.96	5919.91	11839.82	23679.64
G	783.99	1567.98	3135.96	6271.93	12543.85	25087.71
G#/Ab	830.61	1661.22	3322.44	6644.88	13289.75	26579.50
A	880.00	1760.00	3520.00	7040.00	14080.00	28160.00
A#/Bb	932.33	1864.66	3729.31	7458.62	14917.24	29834.48
B	987.77	1975.53	3951.07	7902.13	15804.27	31608.53

Table 4: Notes, rhythm, and MIDI timing - numerical data and parameters.

MIDI note number	Note name	Frequency (Hz)	Period (ms)
21	A0	27.50000	36.363636
22	A#/Bb0	29.13524	34.322702
23	B0	30.86771	32.396317
24	C1	32.70320	30.578051
25	C#/Db1	34.64783	28.861837
26	D1	36.70810	27.241947
27	D#/Eb1	38.89087	25.712974
28	E1	41.20344	24.269816
29	F1	43.65353	22.907655
30	F#/Gb1	46.24930	21.621948
31	G1	48.99943	20.408401
32	G#/Ab1	51.91309	19.262965
33	A1	55.00000	18.181818
34	A#/Bb1	58.27047	17.161351
35	B1	61.73541	16.198159
36	C2	65.40639	15.289026
37	C#/Db2	69.29566	14.430919
38	D2	73.41619	13.620973
39	D#/Eb2	77.78175	12.856487
40	E2	82.40689	12.134908
41	F2	87.30706	11.453828
42	F#/Gb2	92.49861	10.810974
43	G2	97.99886	10.204200
44	G#/Ab2	103.82617	9.631483
45	A2	110.00000	9.090909
46	A#/Bb2	116.54094	8.580676
47	B2	123.47083	8.099079
48	C3	130.81278	7.644513
49	C#/Db3	138.59132	7.215459
50	D3	146.83238	6.810487
51	D#/Eb3	155.56349	6.428243
52	E3	164.81378	6.067454
53	F3	174.61412	5.726914
54	F#/Gb3	184.99721	5.405487
55	G3	195.99772	5.102100
56	G#/Ab3	207.65235	4.815741
57	A3	220.00000	4.545455
58	A#/Bb3	233.08188	4.290338
59	B3	246.94165	4.049540
60	C4	261.62557	3.822256
61	C#/Db4	277.18263	3.607730
62	D4	293.66477	3.405243
63	D#/Eb4	311.12698	3.214122
64	E4	329.62756	3.033727
65	F4	349.22823	2.863457
66	F#/Gb4	369.99442	2.702743

MIDI note number	Note name	Frequency (Hz)	Period (ms)
67	G4	391.99544	2.551050
68	G#/Ab4	415.30470	2.407871
69	A4	440.00000	2.272727
70	A#/Bb4	466.16376	2.145169
71	B4	493.88330	2.024770
72	C5	523.25113	1.911128
73	C#/Db5	554.36526	1.803865
74	D5	587.32954	1.702622
75	D#/Eb5	622.25397	1.607061

Table 5: Notes, rhythm, and MIDI timing - numerical data and parameters.

MIDI note number	Note name	Frequency (Hz)	Period (ms)
76	E5	659.25511	1.516863
77	F5	698.45646	1.431728
78	F#/Gb5	739.98885	1.351372
79	G5	783.99087	1.275525
80	G#/Ab5	830.60940	1.203935
81	A5	880.00000	1.136364
82	A#/Bb5	932.32752	1.072584
83	B5	987.76660	1.012385
84	C6	1046.50226	0.955564
85	C#/Db6	1108.73052	0.901932
86	D6	1174.65907	0.851311
87	D#/Eb6	1244.50793	0.803530
88	E6	1318.51023	0.758432
89	F6	1396.91293	0.715864
90	F#/Gb6	1479.97769	0.675686
91	G6	1567.98174	0.637763
92	G#/Ab6	1661.21879	0.601968
93	A6	1760.00000	0.568182
94	A#/Bb6	1864.65505	0.536292
95	B6	1975.53321	0.506192
96	C7	2093.00452	0.477782
97	C#/Db7	2217.46105	0.450966
98	D7	2349.31814	0.425655
99	D#/Eb7	2489.01587	0.401765
100	E7	2637.02046	0.379216
101	F7	2793.82585	0.357932
102	F#/Gb7	2959.95538	0.337843
103	G7	3135.96349	0.318881
104	G#/Ab7	3322.43758	0.300984
105	A7	3520.00000	0.284091
106	A#/Bb7	3729.31009	0.268146
107	B7	3951.06641	0.253096
108	C8	4186.00904	0.238891
109	C#/Db8	4434.92210	0.225483

MIDI note number	Note name	Frequency (Hz)	Period (ms)
110	D8	4698.63629	0.212828
111	D#/Eb8	4978.03174	0.200883
112	E8	5274.04091	0.189608
113	F8	5587.65170	0.178966
114	F#/Gb8	5919.91076	0.168921
115	G8	6271.92698	0.159441
116	G#/Ab8	6644.87516	0.150492
117	A8	7040.00000	0.142045
118	A#/Bb8	7458.62018	0.134073
119	B8	7902.13282	0.126548
120	C9	8372.01809	0.119446
121	C#/Db9	8869.84419	0.112742
122	D9	9397.27257	0.106414
123	D#/Eb9	9956.06348	0.100441
124	E9	10548.08182	0.094804
125	F9	11175.30341	0.089483
126	F#/Gb9	11839.82153	0.084461
127	G9	12543.85395	0.079720

Table 6: Notes, rhythm, and MIDI timing - numerical data and parameters.

Message Type	Channel	Parameter	Time
program change _	0	program=71	0
control change _	0	control=10, value=64	0
note on _	0	note=73, velocity=96	0
note off _	0	note=73, velocity=0	96
note on _	0	note=83, velocity=96	0
note off _	0	note=83, velocity=0	288
note on _	0	note=80, velocity=96	0
note off _	0	note=80, velocity=0	480
note on _	0	note=78, velocity=96	0
note on _	0	note=76, velocity=96	192
note off _	0	note=78, velocity=0	0
note off _	0	note=76, velocity=0	96
note on _	0	note=75, velocity=96	0

3 Oscillator synthesis

The mandatory synthesis stage combines a periodic source with an amplitude envelope. Implement sine, sawtooth, triangle, and square waveforms. Parameters include waveform, phase, fundamental frequency, volume, and detuning; optional oscillator filtering uses a VCF. Phase and time delay are related for an individual sinusoid but are not interchangeable parameters for an arbitrary waveform. A Gaussian-noise source may be added; its amplitude is adjustable, but it has no unique fundamental or phase. Combine several oscillators as a weighted sum. Detuning uses nearby frequency offsets and weights: x_detuned(t) = sum_i A_i*x(2*pi*(f_b+f_i)*t). The assignment permits a justified choice of oscillator count, offsets, and weights as functions of the detune control. Preserve amplitude headroom when mixing voices. Naively sampled nonsinusoidal waveforms can alias; discuss this limitation rather than treating every generated high harmonic as physically meaningful.

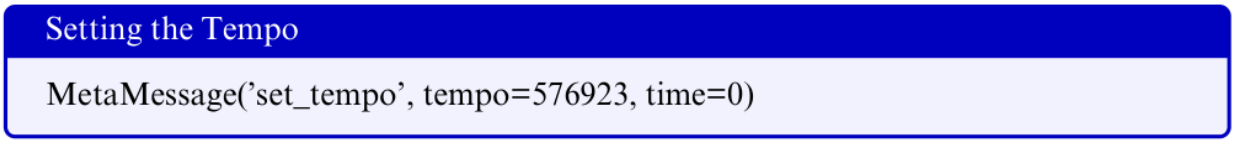


Figure 2: Diagram or recorded result: Oscillator synthesis.

4 AHDSR envelope

Implement adjustable Attack, Hold, Decay, Sustain, and Release. Attack rises to peak, Hold stays at peak, Decay falls toward the sustain level, Sustain continues until note release, and Release falls to silence after note-off. Use exponential-like transitions as requested. A normalized exponential segment must have explicit endpoint behavior; the illustrative piecewise exponentials in the assignment should not be copied with discontinuities or without amplitude offsets. Support zero Hold and the specified zero-Sustain behavior. Velocity controls a documented amplitude mapping. Release must begin from the current envelope level when a short note ends before attack/decay has finished. Preserve the original envelope diagrams, including the exponential example.

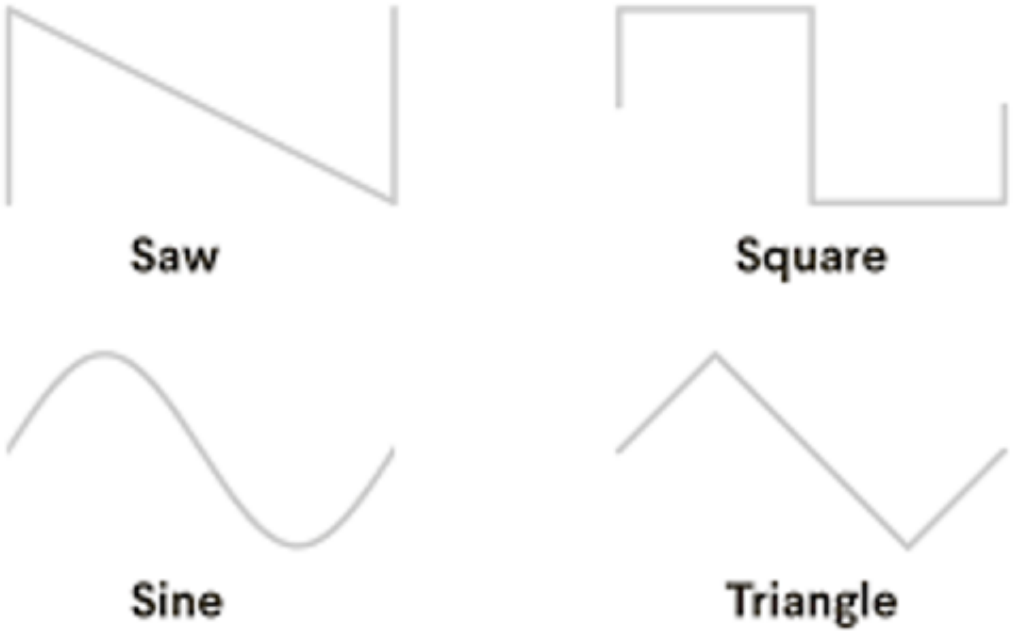


Figure 3: Diagram or recorded result: AHDSR envelope.

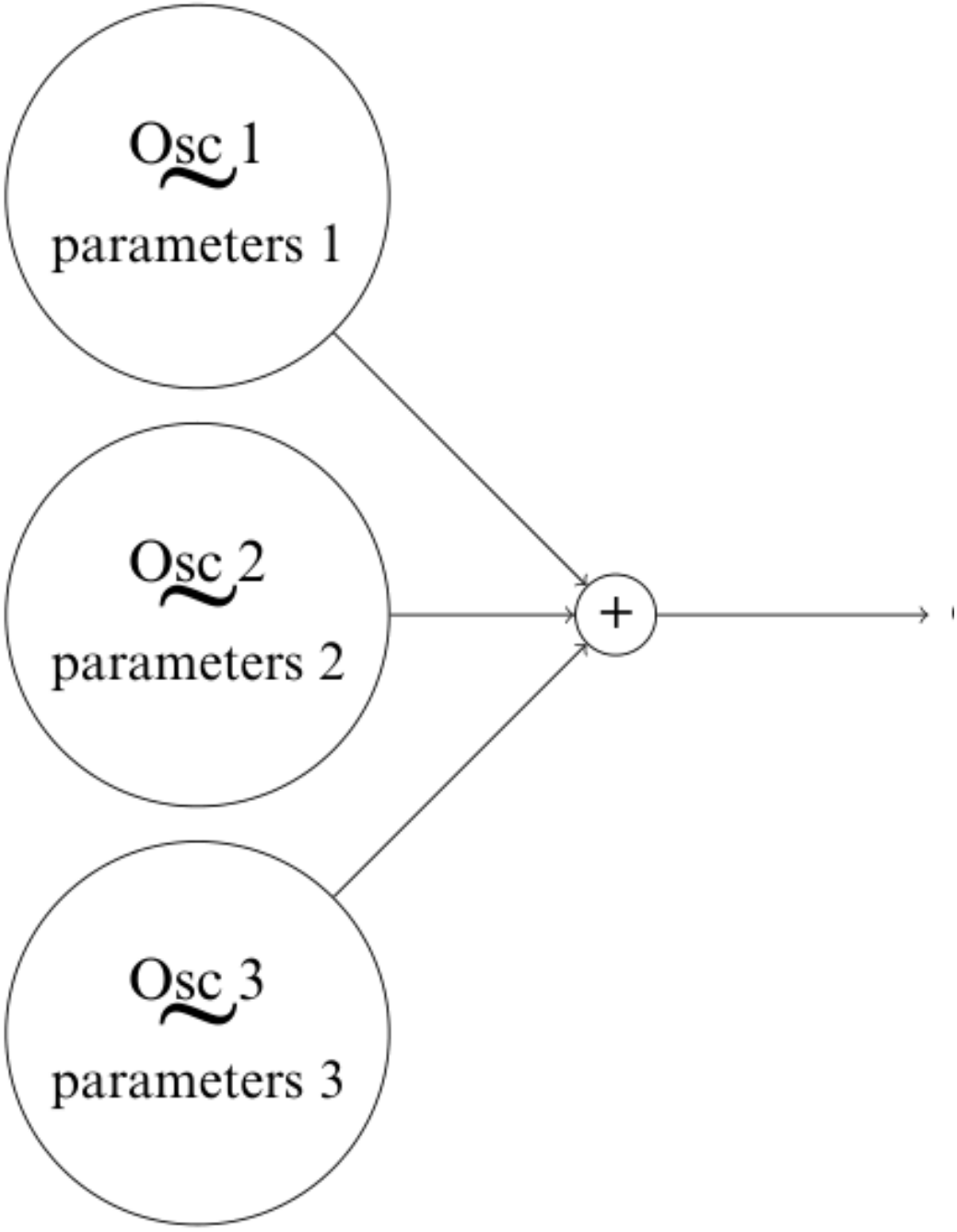


Figure 4: Diagram or recorded result: AHDSR envelope.

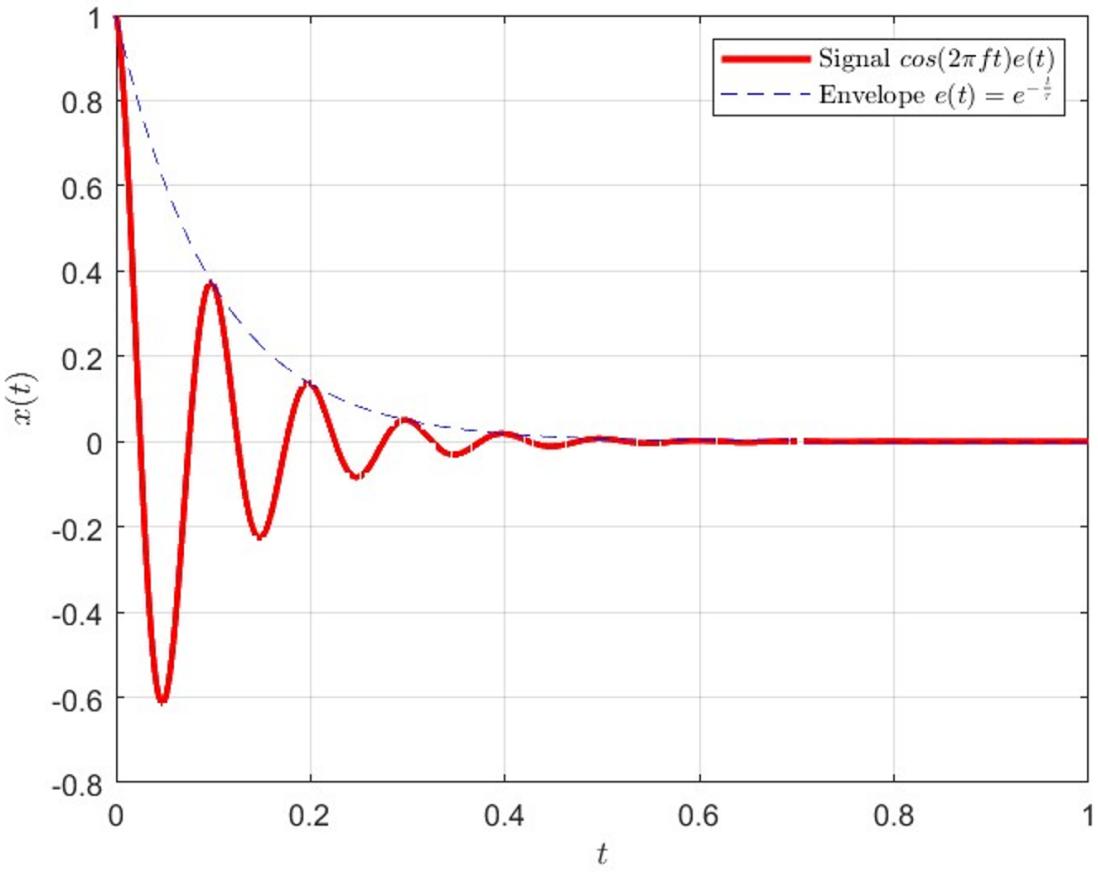


Figure 5: Diagram or recorded result: AHDSR envelope.

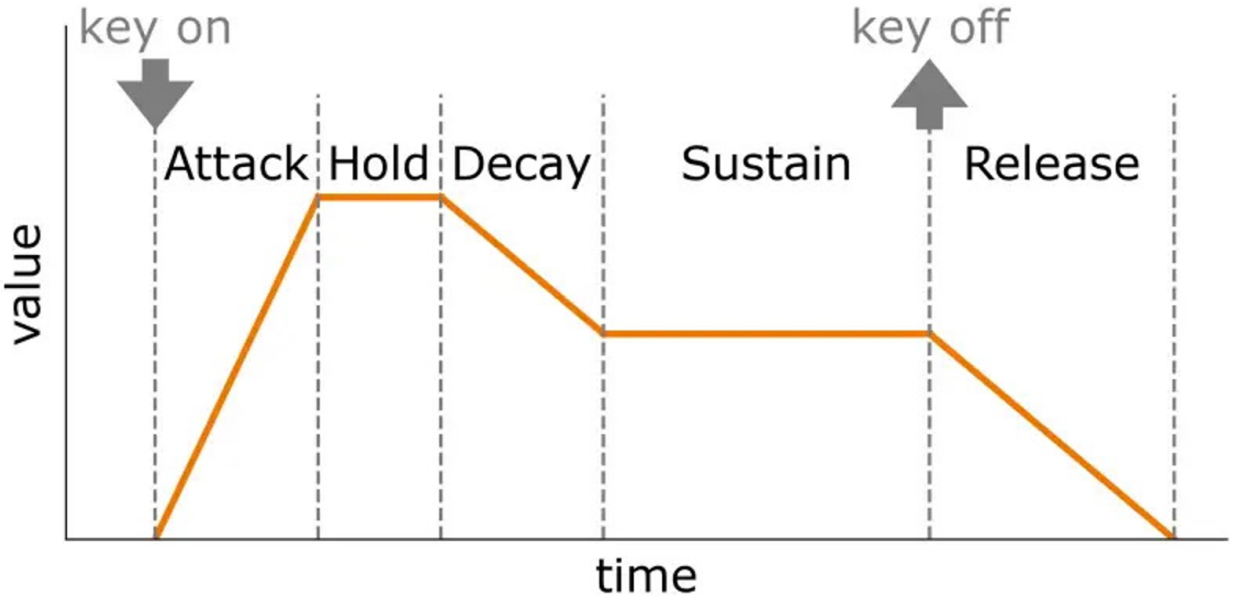


Figure 6: Diagram or recorded result: AHDSR envelope.

5 Shared effect interface

An effect receives audio and returns transformed audio. A rack cascades several effects in an ordered chain. Every effect has input gain, output gain, and a wet/dry parameter w from zero to one; mix as (1-w)*dry + w*wet, with defined gain placement. Order matters for nonlinear effects. Expose parameters and sample rate rather than embedding unexplained constants.

Effect	Main points	Bonus points
Phaser	0.05	0.05
Flanger	0.05	0.05
Chorus	0.05	0.05
Delay	0.10	0.10
Reverberation	0.20	none
Pitch shifter	0.40	none
Vocoder	0.30	none
Saturation	0.25	none
Compressor	0.25	none
Voltage-controlled filter	0.10	none

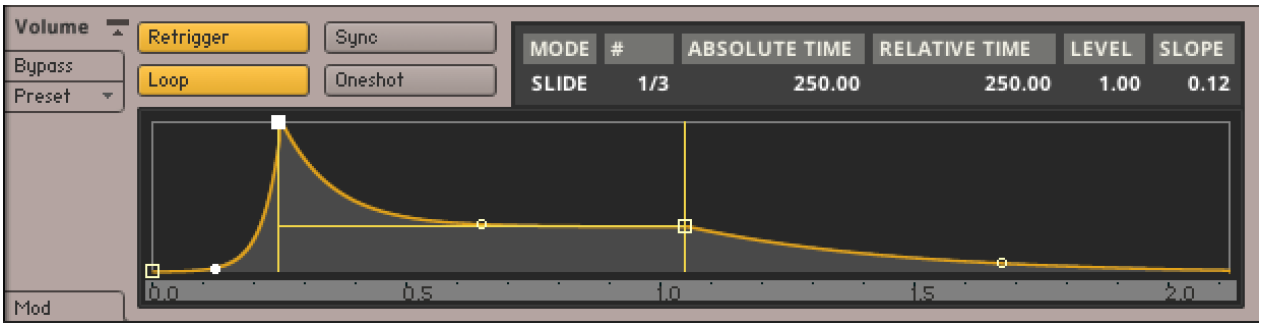


Figure 7: Diagram or recorded result: Shared effect interface.

Table 8: Shared effect interface - numerical data and parameters.

Effect	Point	Bonus
Phaser	0.05	0.05
Flanger	0.05	0.05
Chorus	0.05	0.05
Delay	0.1	0.1
Reverberation	0.2	-
Pitch Shifter	0.4	-
Vocoder	0.3	-
Saturation	0.25	-
Compressor	0.25	-
Voltage-Controlled Filter	0.1	-

6 Saturation and compression

Implement hard clipping to [-T,T], soft clipping T*tanh(x/T), and the specified variable saturation T*abs(x/T)^a*sign(x). Drive D and bias b change the input to f(D*x+b). Define $T > 0$ and parameter ranges. This variable-power law is not necessarily a bounded clipper outside its intended range. Nonlinear processing can generate harmonics and aliasing. The compressor has threshold, ratio, attack, and release controls. Above a threshold in decibels, the steady-state level is T+(L-T)/R. Convert the resulting gain reduction back to a linear multiplier. Do not mix linear amplitudes with decibel thresholds or interpret a compression ratio as a fixed number of decibels. Use an appropriate level/envelope detector and discretize attack/release behavior. The source suggests averaging recent samples for smoother response; specify whether the detector uses magnitude, RMS, or another quantity.

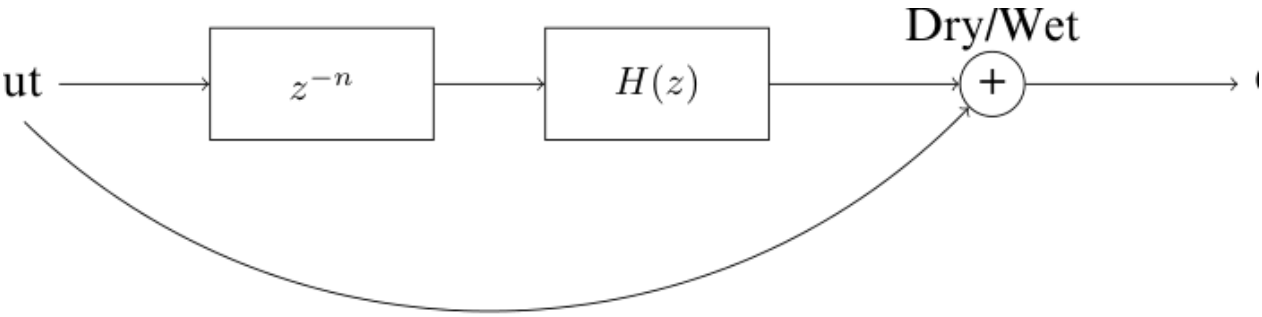


Figure 8: Diagram or recorded result: Saturation and compression.

7 Delay, phaser, flanger, and chorus

Delay uses z^(-n) and an optional H(z) in the delayed branch. Convert time delay to samples using the sample rate. H(z)=1 gives the simple echo path; filtering or low-frequency modulation can add the stated extensions. Phaser mixes a dry signal with a cascade of all-pass-filter outputs. Phase differences produce frequency-dependent cancellation and peaks. Flanger uses a short delayed branch, optionally with feedback and modulation. Chorus uses multiple differently delayed/modulated branches to suggest several voices. Preserve the source diagrams and distinguish these related structures. The original typical-setting tables list phasing 0-3 ms with a little feedback, large modulation width, and slow speed; flanging 3-13 ms with substantial feedback and medium width/speed; chorus 13-30 ms with no feedback, small width, and medium speed. Earlier prose gives a 5-10 ms starting range for flanger/chorus. These are illustrative settings, not universal definitions. Optional LFO modulation applies to the three effects.



Figure 9: Diagram or recorded result: Delay, phaser, flanger, and chorus.

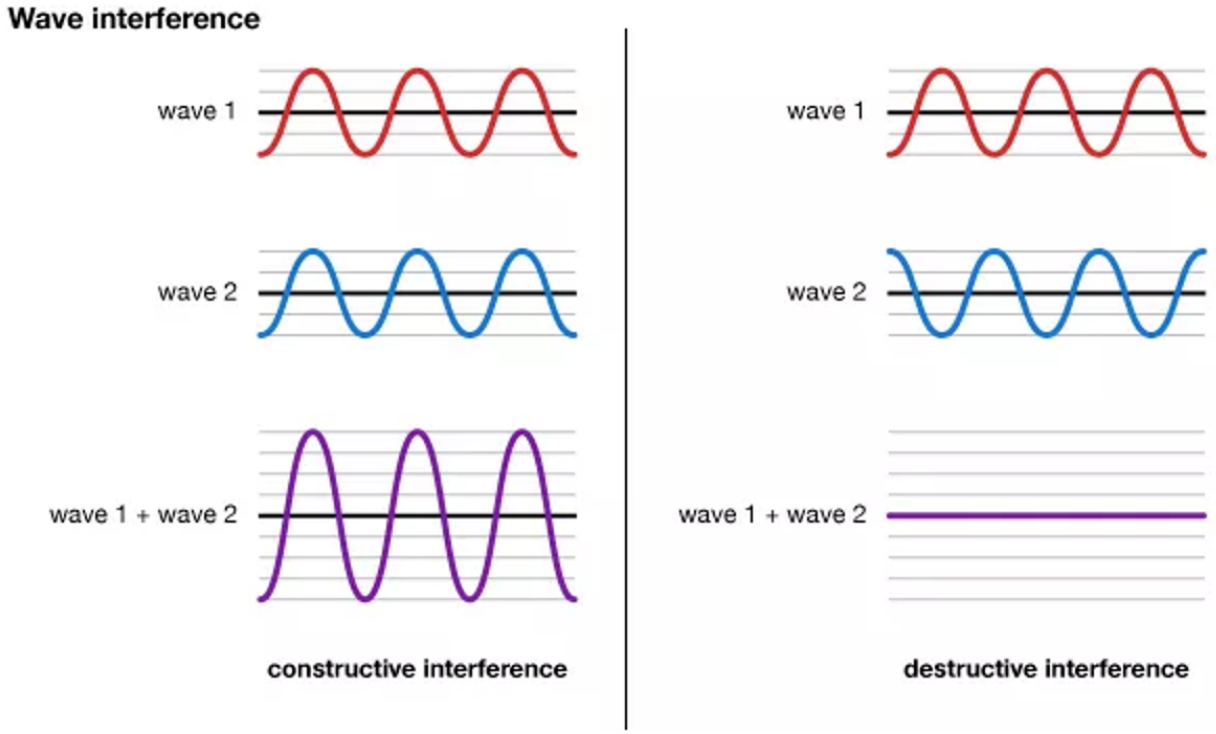


Figure 10: Diagram or recorded result: Delay, phaser, flanger, and chorus.

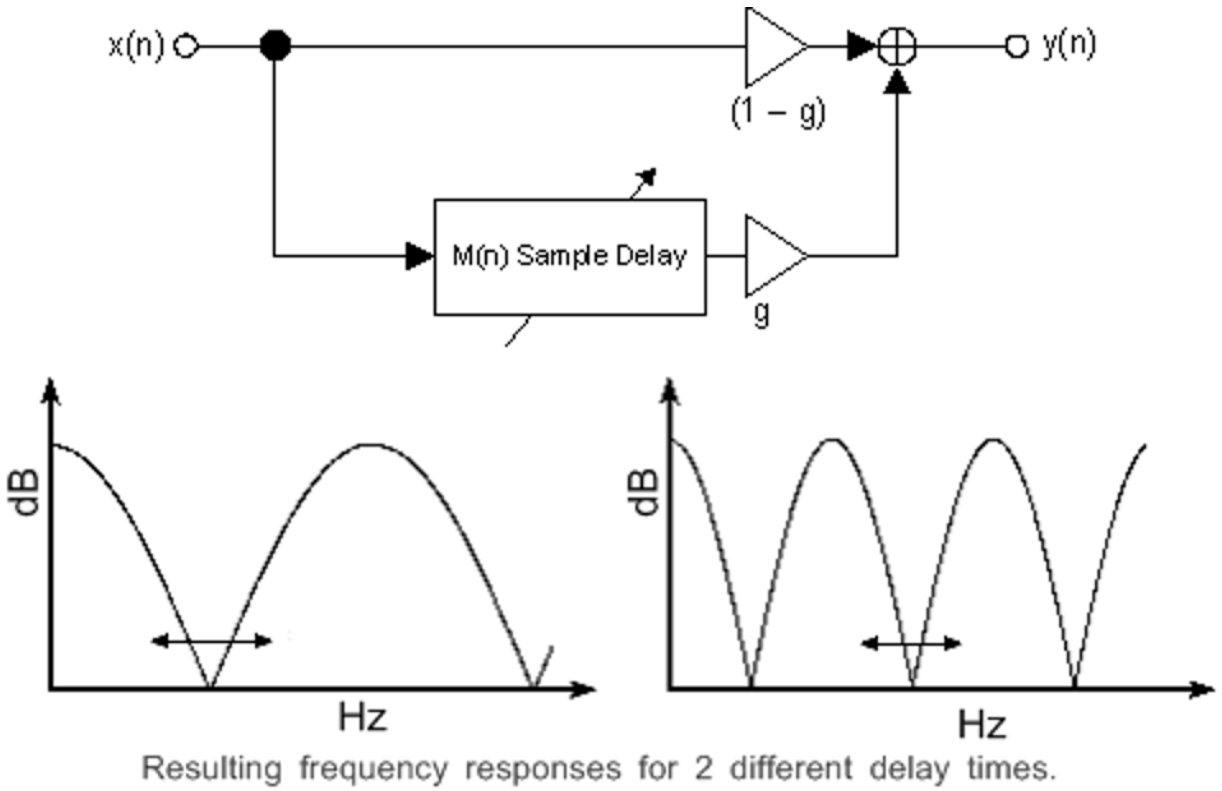


Figure 11: Diagram or recorded result: Delay, phaser, flanger, and chorus.

Table 9: Delay, phaser, flanger, and chorus - numerical data and parameters.

Effect	**Delay time**	**Feedback**	**Modulation**	
			Width	Speed
Phasing	0-3ms	A little	Lots	Slow

Table 10: Delay, phaser, flanger, and chorus - numerical data and parameters.

Effect	**Delay time**	**Feedback**	**Modulation**	
			Width	Speed
Flanging	3-13ms	Lots	Medium	Medium

Table 11: Delay, phaser, flanger, and chorus - numerical data and parameters.

Effect	**Delay time**	**Feedback**	**Modulation**	
			Width	Speed
Chorus	13-30ms	None	A little	Medium

8 Reverberation and filters

Digital reverb can combine delayed branches with gains, using relatively prime delay lengths as suggested. A realistic sustained decay needs a defined network or sufficiently rich finite response; three isolated feedforward delays alone produce only a few echoes. Convolution reverb filters audio with a measured room impulse response. Match sample rates, manage output length, and document the impulse-response source. The VCF exposes low-pass, high-pass, band-pass, or band-stop behavior and adjustable cutoff. IIR implementations include direct forms I, II, and transposed structures. Keep coefficients and filter state separate. Direct form II stores internal state rather than merely a list of raw recent input samples. An LFO below approximately 20 Hz can vary oscillator frequency or filter cutoff. Recompute coefficients from a stable filter design; arbitrary coefficient perturbation may make poles unstable. Vibrato is frequency modulation and tremolo is amplitude modulation.

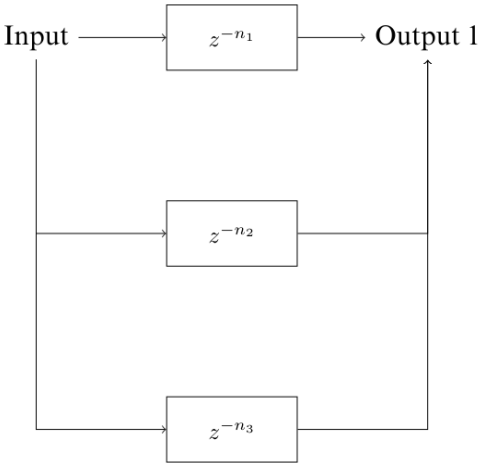


Figure 12: Diagram or recorded result: Reverberation and filters.

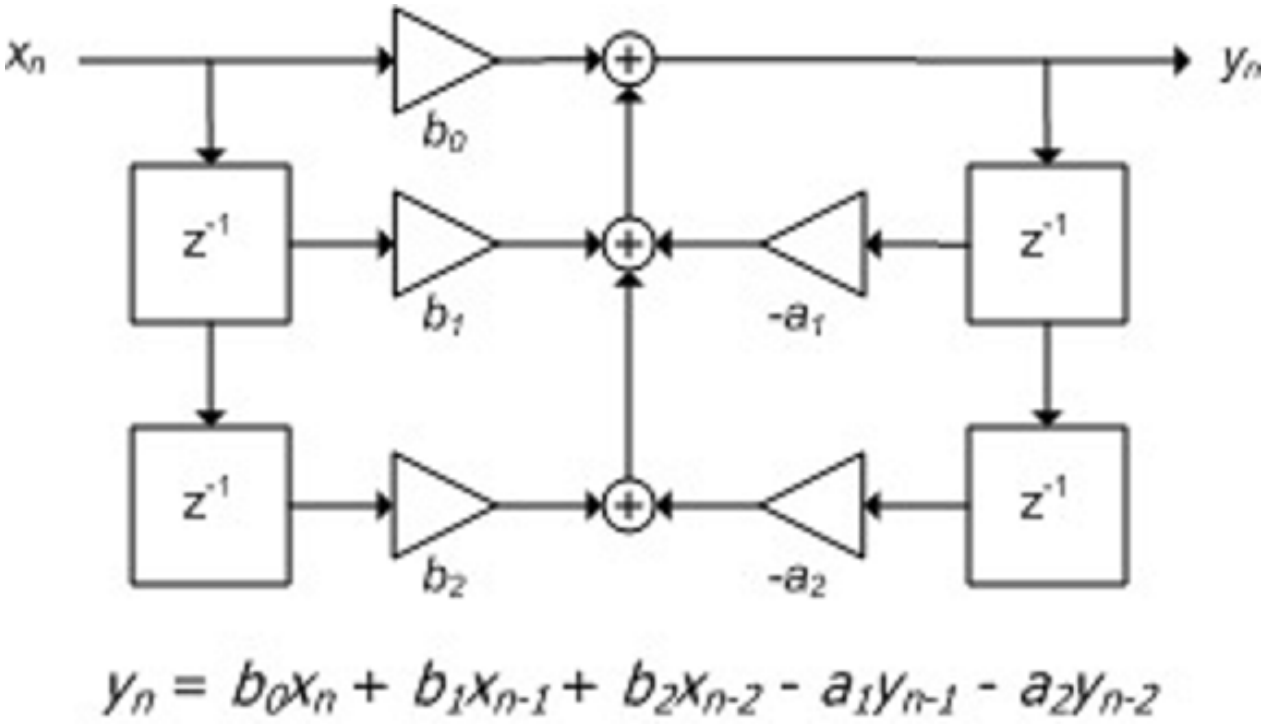


Figure 13: Diagram or recorded result: Reverberation and filters.

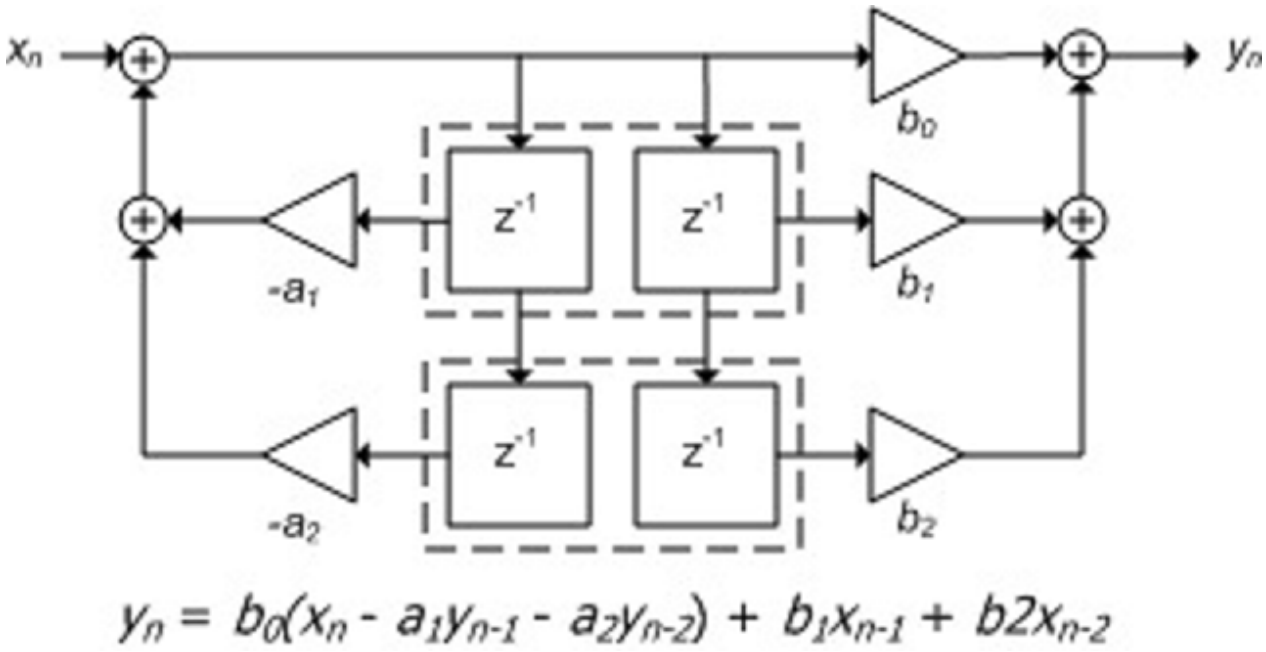


Figure 14: Diagram or recorded result: Reverberation and filters.

9 Vocoder and pitch shifting

For the vocoder, divide the useful spectrum into logarithmically spaced bands, filter the carrier and speech modulator in each band, estimate modulator envelopes, apply each envelope to its carrier band, and sum. The assignment gives y[n] = sum_i abs(m_i[n])*c_i[n]; practical smoothing of the rectified envelope improves behavior. Use roughly a dozen or more bands, stable filters, and a useful lower frequency bound. Preserve carrier duration by padding or truncating the modulator. For pitch shifting, a three-semitone shift maps C-sharp4 to E4 and uses frequency ratio 2^(3/12), not a literal ratio of three. Simple resampling changes duration. The proposed phase-vocoder method first changes time scale without changing pitch, then resamples to restore duration. Analysis uses STFT, processing tracks/corrects phase with optional phase locking, and synthesis uses inverse STFT and overlap-add with compatible windows/hops. Preserve the original phase-vocoder diagram and explain the chosen implementation's artifacts and limits.

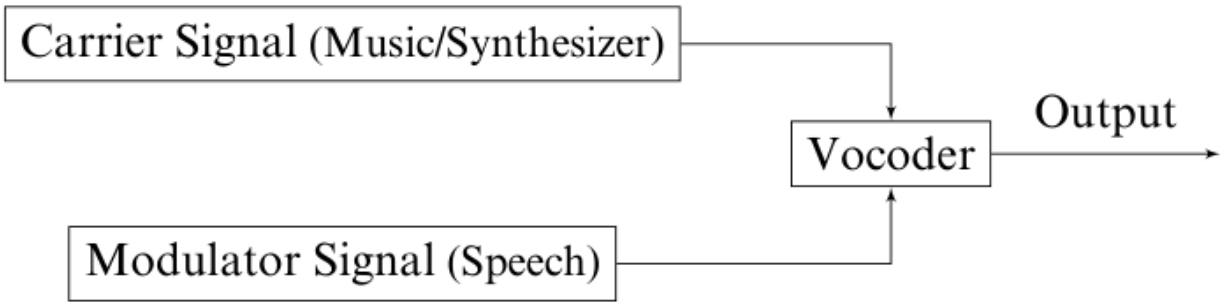


Figure 15: Diagram or recorded result: Vocoder and pitch shifting.

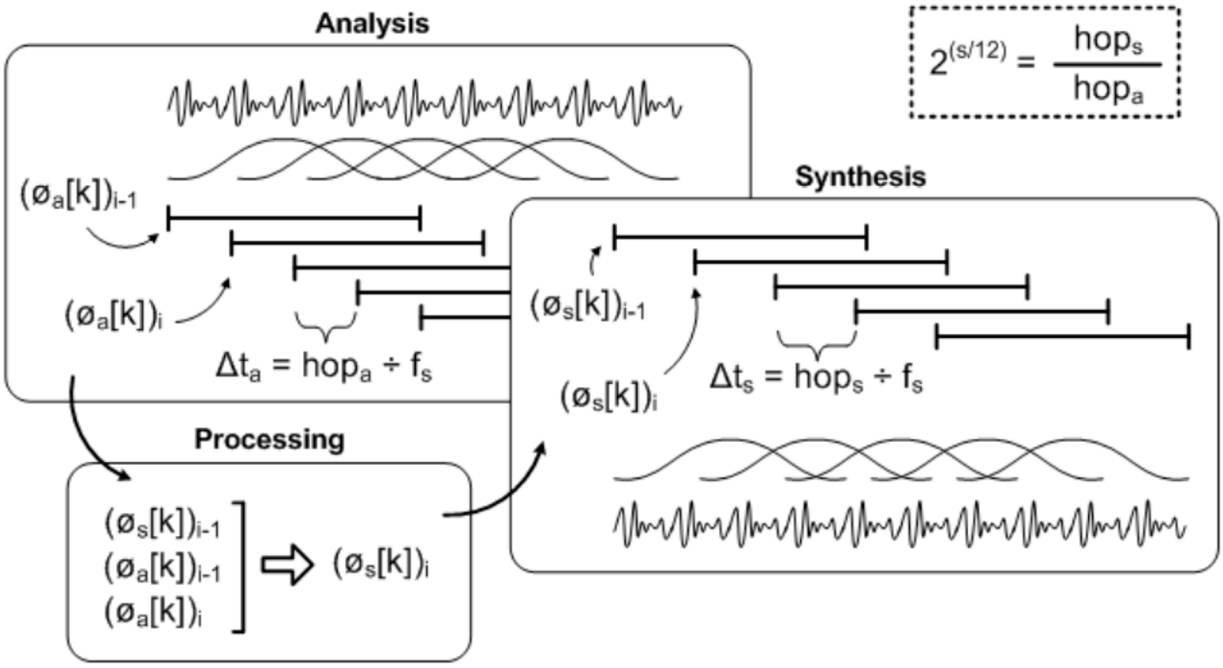


Figure 16: Diagram or recorded result: Vocoder and pitch shifting.